Catch stockouts before the shelf goes empty

Inventory Risk Radar readout for Harvest Lane Markets

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54% more stockouts caught for the same planner effort

- 62% of radar alerts were real stockouts, vs 20% with today's cover rule
- Same workload: 732 real stockouts caught vs 475 (+54%)
- 78% fewer alerts to triage than today's rule (1,172 vs 5,372)
- Forecast error down 14%, and 27% on promotion days
- Ask: an 8-week pilot in 2 regions to set targets on your baseline

Stockouts are costly and concentrated

4.2% of shelves

Share of store/SKU days that end with zero on hand, last 12 months

19% on promo days

Stockouts are 4.6x more likely when you are paying to drive demand

$10.8M a year

Estimated lost sales before substitution, on the 400-SKU slice modeled

Why planners find out too late

How it works today

Sales, inventory, POs, promos and lead times sit in separate reports

Days-of-cover rule ignores promotions and late suppliers

5,372 alerts a day, 80% of them false

What planners need

One ranked list of what will run out before replenishment

The reason in units, and a suggested action

A place to decide and record it

Value for the sponsor and the planning team

VP Supply Chain

Higher on-shelf availability where it matters most: promotions and holidays

Lost sales measured and tracked as one KPI

Supplier issues visible the day they start

Replenishment planning lead

A morning queue ranked by risk and revenue

78% fewer alerts, 3x more of them real

Expedite, transfer or reorder suggested; planner decides

How the Risk Radar works

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Every morning, raw feeds become a ranked decision

1. Ingest: Lakeflow pipelines land POS, inventory, PO, promo and supplier feeds and drop bad records
2. Govern: Unity Catalog holds every table, model and KPI with access rules and lineage
3. Predict: a 28-day forecast per store/SKU, then a transparent stockout-risk score with drivers
4. Act: planners work the queue in an app on Lakebase; Genie and a dashboard answer the rest

This morning's queue: top risks with the reason and action

Item / store	Out in	Why	Suggested action
Green Beans 2ct / Stonebrook	1 day	Supplier 348 units late	Expedite 348 from Riverbend Fresh
Heavy Cream / Oakhurst	1 day	Supplier late + promo	Expedite 138 from Summit Dairy
Sweet Potatoes / Yarrow	1 day	Promo +246 units vs usual	Expedite 192 from Pacific Fresh
2% Milk Organic / Juniper Bay	2 days	Dairy supplier 58% on time	Expedite 84 from Summit Dairy
Peanut Butter / Stonebrook	12 days	Promo demand 2x usual	Reorder 96 from Summit Pantry

Two late suppliers explain most of today's risk

Late supplier delivery

2,094 of 2,338 alerts

One call to Summit Dairy covers 516 store/SKUs

Upcoming promotion

184 alerts

Forecast 27% more accurate on promo days

Stock cover gaps

60 alerts

No open PO or a demand surge

Measured against today's method

KPI	Today's method	Risk Radar
Alert precision (4-week backtest)	20%	62%
Real stockouts caught, same workload	475	732 (+54%)
Alerts to triage	5,372	1,172 (-78%)
Forecast error (WAPE), all days	0.53	0.45 (-14%)
Forecast error (WAPE), promo days	0.55	0.40 (-27%)

Sizing the value on your baseline

- Lost sales today: about $10.8M a year on the 400 SKUs modeled, before substitution
- Assume half is recovered by substitution: $5.4M at stake
- Radar catches 38% of stockouts; if planners prevent half of those, about $1.0M a year in sales is recovered
- Planner time: 78% fewer alerts to review every morning
- Pilot sets the real targets: precision, stockout rate, time to action

Built to run in production

Governed and secure

Unity Catalog: lineage, tags, region row filters, masked financials

App runs as its own service principal; short-lived tokens, no passwords

Synthetic data only; nothing leaves your account

Reliable and portable

Data-quality rules drop bad records before they reach a forecast

Model versioned in MLflow; @champion alias, easy rollback

One bundle deploys pipeline, job, model, Lakebase, Genie, app

An 8-week pilot sets targets on your data

- Weeks 1-2: connect POS, inventory, PO, promo and lead-time feeds for 2 regions; measure the baseline
- Weeks 3-4: run the radar in shadow mode and compare alerts with real stockouts
- Weeks 5-8: planners work the queue in the app; track precision, stockout rate, time to action
- Week 8: decision gate on targets and roll-out to all regions

Decisions we need to start

- Sponsor: approve the 8-week, 2-region pilot
- Planning lead: nominate 2 pilot planners and agree on the alert tiers
- IT / data: grant read access to the 5 source feeds
- Together: agree on success metrics and the week-8 decision gate



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